ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

MATHEMATICS

PRACTICE EXAM 2

MARKS: 100 **TIME: 2 hours**

Covers the full Grade 7 year: whole numbers, ratio, rate and money, exponents, fractions and decimals, integers, patterns and algebra, geometry, transformations, 3D objects, measurement, data handling and probability.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of NINE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL your working. Answers alone may not get full marks.
4. Calculators may NOT be used.
5. Diagrams are NOT necessarily drawn to scale.
6. Write neatly and legibly.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 4^3 = …

A 12 B 64 C 7 D 81

1.2 Which number is a multiple of both 4 and 6?

A 16 B 18 C 24 D 30

1.3 $\frac{3}{5}$ written as a percentage is …

A 35% B 60% C 53% D 30%

1.4 0,4 × 100 = …

A 0,004 B 4 C 40 D 400

1.5 −3 − 4 = …

A −7 B 1 C −1 D 7

1.6 How many faces does a triangular prism have?

A 3 B 5 C 6 D 9

1.7 The angles of any quadrilateral add up to …

A 90° B 180° C 270° D 360°

1.8 1 m² = … cm²

A 100 B 1 000 C 10 000 D 1 000 000

1.9 If $2x = 18$, then x = …

A 16 B 9 C 20 D 36

1.10 The probability of rolling a 6 with one fair dice is …

A $\frac{1}{6}$ B $\frac{1}{2}$ C 6 D $\frac{1}{3}$

[10]

QUESTION 2: WHOLE NUMBERS, RATIO, RATE AND MONEY

2.1 Use the distributive property to calculate 8 × 99. Show your method. (2)

2.2 Lerato and Sipho share R720 in the ratio 5 : 3. How much does each person get? (3)

2.3 A taxi travels 240 km in 3 hours. Calculate its average speed in km/h. (2)

2.4 A shop buys a radio for R350 and sells it for R490. Calculate: (a) the profit; (b) the profit as a percentage of the cost price. (3)

2.5 A bicycle costs R800 before VAT. VAT is 15%. Calculate the price including VAT. (2)

2.6 Nomsa invests R2 000 at 8% simple interest per year. Calculate the interest she earns after 3 years and the total amount she will then have. (3)

[15]

QUESTION 3: EXPONENTS

3.1 Calculate: (a) 2^5 (b) $(3 + 2)^2 - \sqrt{81}$ (c) 4×10^3 (4)

3.2 Which is larger, 3^4 or 4^3? Show how you decided. (2)

3.3 Write 360 as a product of prime factors. Give your answer in exponential form. (2)

[8]

QUESTION 4: FRACTIONS AND DECIMALS

4.1 Calculate. Give your answers in the simplest form.

4.1.1 $\frac{5}{6} - \frac{1}{3}$ (2)

4.1.2 $2\frac{2}{5} \times 1\frac{1}{4}$ (3)

4.1.3 $\frac{1}{2} + \frac{2}{5} + \frac{3}{10}$ (2)

4.2 Write 0,125 as a common fraction in its simplest form. (2)

4.3 Calculate: (a) 12,5 × 0,3 (b) 9,36 ÷ 4 (3)

4.4 There are 32 learners in a class. $\frac{3}{8}$ of them walk to school. How many learners walk to school? (2)

[14]

QUESTION 5: INTEGERS

5.1 Write < or > between each pair of numbers: (a) −9 … −4 (b) −1 … −10 (2)

5.2 Calculate: (a) −6 + (−8) (b) 4 − 11 (c) −20 + 35 − 5 (3)

5.3 A diver is 18 m below sea level. She swims 7 m upwards. Write her new position as an integer. (1)

[6]

QUESTION 6: PATTERNS, FUNCTIONS AND ALGEBRA

6.1 Look at the number pattern: 2 ; 6 ; 10 ; 14 ; …

6.1.1 Write down the next two terms. (2)

6.1.2 Write down the rule for the *n*th term. (2)

6.1.3 Calculate the 50th term. (1)

6.2 Write down the next term of the pattern 3 ; 6 ; 12 ; 24 ; … and describe the rule in words. (2)

6.3 Study the table.

x	1	2	3	4	10
y	7	11	15	19	?

6.3.1 Write down the rule in the form $y = \ldots$ (2)

6.3.2 Find the value of y when $x = 10$. (1)

6.4 I think of a number, multiply it by 5 and add 4. The answer is 39. Write an equation and solve it to find the number. (3)

6.5 Calculate the value of $3m^2$ if $m = 2$. (2)

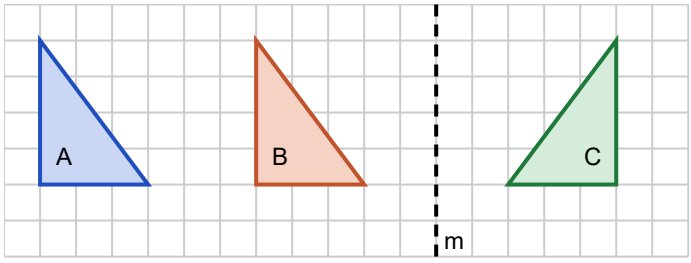
[15]

QUESTION 7: GEOMETRY, TRANSFORMATIONS AND 3D OBJECTS

7.1 Classify each triangle according to its sides AND its angles: (a) sides 5 cm, 5 cm and 8 cm with an obtuse angle; (b) angles 90°, 45° and 45°. (2)

7.2 Write down TWO properties of a parallelogram. (2)

7.3 On the grid below, triangle B is the image of triangle A, and triangle C is the image of triangle B. Line m is a mirror line. Describe each transformation fully.



Each square on the grid is 1 unit by 1 unit.

7.3.1 A → B (2)

7.3.2 B → C (2)

7.4 A square-based pyramid has how many (a) faces, (b) vertices and (c) edges? (3)

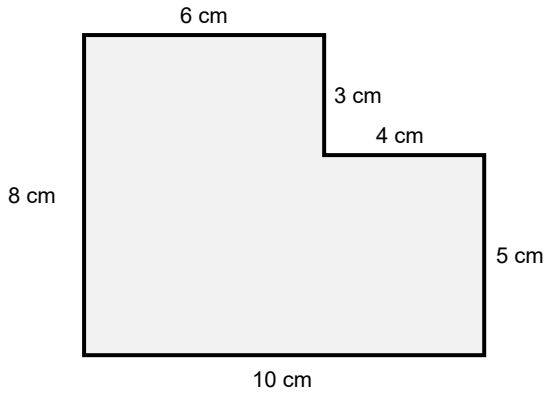
7.5 Name the 3D object whose net is made of six identical squares. (1)

[12]

QUESTION 8: MEASUREMENT

8.1 Convert: (a) 4,5 cm to mm (b) 3 500 m to km (c) 750 g to kg (3)

8.2 The diagram shows the floor plan of an L-shaped room.



8.2.1 Calculate the area of the floor. (3)

8.2.2 Calculate the perimeter of the floor. (2)

8.3 A cube has sides of 6 cm. (a) Calculate its volume. (b) How many small cubes with sides of 2 cm would fit exactly inside it? (4)

[12]

QUESTION 9: DATA HANDLING AND PROBABILITY

9.1 40 learners were asked to name their favourite sport. The results are shown in the table.

Sport	**Soccer**	**Netball**	**Rugby**	**Cricket**	**Athletics**
Number of learners	15	10	6	5	4

9.1.1 Which sport is the mode? (1)

9.1.2 What percentage of the learners chose netball? (2)

9.1.3 The data is shown on a pie chart. Calculate the size of the angle of the sector for soccer. (2)

9.2 Two coins are tossed at the same time. (a) List all the possible outcomes. (b) What is the probability of getting two heads? (3)

[8]

TOTAL: 100

END OF QUESTION PAPER